UESTC4003 - Control Engineering

Design Exercise

Background

The increasing integration of renewable energy sources into electrical power networks presents significant operational challenges due to the intermittent nature of generation. Wind power generation, whilst environmentally sustainable, can introduce fluctuations in power output that may adversely affect grid stability and power quality. Energy storage technologies can help mitigate these fluctuations and improve the reliability of renewable energy systems.

You are employed as a Control Systems Engineer within an engineering consultancy company. A grid operator is planning a significant expansion of wind generation capacity and is considering investment in advanced energy storage technologies to improve the stability and reliability of the local electrical network.

The client currently operates:

- A wind generation system
- A battery energy storage system

The client is proposing to install a supercapacitor energy storage system to provide rapid response during transient conditions and sudden changes in demand.

Project Objectives

The objectives of this study are to:

Technical Objectives

- Develop an understanding of the role of energy storage technologies in supporting renewable energy integration.
- Investigate the operational characteristics of batteries and supercapacitors for grid support applications.
- Develop MATLAB/Simulink models representing:
    - Wind generation
    - Battery energy storage
    - Supercapacitor energy storage
    - An integrated wind-battery-supercapacitor system
- Evaluate system performance under a variety of operating conditions.

Engineering Evaluation Objectives

- Critically evaluate the suitability of supercapacitors for transient power support.
- Assess the limitations of the proposed storage architecture.
- Compare alternative energy storage technologies.
- Investigate the impact of uncertainty and operating disturbances.
- Develop evidence-based engineering recommendations.

Design Specification

The proposed system must satisfy the following client requirements:

Functional Requirements

The system should:

- Maintain a stable electrical power supply.
- Compensate for fluctuations in wind generation.
- Meet load demand during transient disturbances.
- Maintain acceptable system performance under varying operating conditions.

Performance Requirements

The system should:

- Recover from disturbances as rapidly as possible.
- Minimise power deviations from the desired load demand.
- Limit unnecessary energy storage utilisation.
- Demonstrate stable operation throughout the simulation period.

Design Constraints

The client wishes to:

- Minimise the required supercapacitor capacity.
- Minimise battery stress.
- Minimise overall system complexity.
- Maintain acceptable performance under uncertainty.

System Modelling

Develop MATLAB/Simulink models representing:

1. Wind Generation System: Use the provided wind generation dataset (gridwatch_wind_data.csv) covering a 24-hour period.

   You may make reasonable engineering assumptions where necessary. All assumptions must be clearly stated and justified.

2. Battery Energy Storage Model: Develop an appropriate battery model capable of: compensating for slow variations in power generation; Maintaining long-term energy balance.

   The battery parameters should be justified using engineering reasoning.

3. Supercapacitor Model: Develop a MATLAB/Simulink representation of a supercapacitor system suitable for: Fast transient support, Rapid response to sudden load changes.

   All design assumptions should be justified.

4. Integrated System Model: Develop a complete integrated model consisting of: Wind generation, Battery storage, Supercapacitor storage, Load demand

Performance Investigation

<u>Case Study 1 – Constant Load Operation</u>

In this investigation, the performance of the proposed wind-battery-supercapacitor system should be evaluated under normal operating conditions. Assume a constant electrical load demand of 480 kW over the 24-hour simulation period. Students should assess the ability of the battery energy storage system to compensate for fluctuations in wind generation and maintain a stable power supply to the

load. Particular attention should be given to energy balance, battery utilisation, and overall system stability. Based on the simulation results, students should evaluate whether a battery-only solution is sufficient under normal operating conditions and discuss the operational benefits and limitations of the storage configuration adopted.

Case Study 2 – Disturbance Response Investigation

The second investigation should examine the ability of the integrated system to respond to a sudden and short-duration increase in load demand. Students should assume that the system initially operates with a constant load of 480 kW, before experiencing a temporary increase to 720 kW for a period of 10 minutes at midday. The performance of both the battery and supercapacitor should be analysed with respect to response time, power-sharing capability, system stability, and recovery characteristics. Students should critically evaluate the effectiveness of the supercapacitor in supporting the battery during transient conditions and determine whether the proposed design satisfies the client's requirement for rapid disturbance rejection while maintaining acceptable system performance.

Case Study 3 – Severe Operating Scenario

Students should develop and justify an additional operating scenario that represents a significant challenge to the power system. Examples may include a sudden reduction in wind generation, multiple load disturbances, an extended period of peak demand, or a combined generation and load event. The purpose of this investigation is to assess the resilience and robustness of the proposed storage architecture under abnormal operating conditions. Students should justify the choice of scenario, analyse the response of all system components, identify any performance limitations, and evaluate the extent to which the proposed energy storage strategy can maintain system stability and supply security. Recommendations for improving the design should be provided where weaknesses are identified.

Engineering Recommendation to the Client

Drawing upon the results of all investigations, students should provide a professional engineering recommendation regarding the suitability of integrating a supercapacitor into the wind-battery system. The recommendation should be supported by evidence obtained from the simulations and should consider system performance, robustness, operational benefits, limitations, and practical implementation issues. Students should clearly state whether the investment in a supercapacitor is justified and explain why this represents the most appropriate solution for the client. Particular emphasis should be placed on the trade-offs between technical performance, system complexity, and operational reliability.

Report Structure

The report should be written in the style of a professional technical report and should document the modelling approach, assumptions, simulation methodology, results and analyses associated with the three performance investigations. Students should include appropriate figures, plots and simulation outputs, and demonstrate an understanding of system behaviour by interpreting the results obtained under each case study. The report should compare the system performance under normal and disturbed operating conditions, evaluate the influence of key design parameters, and discuss the limitations of the modelling approach where appropriate. Conclusions should be supported by evidence from the simulation results, with all assumptions and design choices clearly justified. Relevant literature and technical sources should be used to support the discussion and analysis.

You are expected to use 10+ refereed sources of information, such as scientific journal articles, technical reports and books to support your report (*You may wish to use University of Glasgow library search for resources https://gla.on.worldcat.org/advancedsearch*).

Your report should be a pdf file that must be submitted to Moodle via the assessment link. Failure to submit the report correctly may result in a mark of zero or a CW, depending on the circumstances.

Deadline of the report submission: **14th of December 2026** ***(Beijing time)***

Word limit: about 1200 words (plus reference list)

Total mark: 15% (assuming total mark of the course as 100%)

Late submissions will not be accepted.

Assessment matrix is shown below.

	Maximum total mark	Marks
Demonstrate understanding of the background: Has the student demonstrated an understanding of the importance of storage devices to control the stability of power grid?	20%	
Technical Understanding: Has the student demonstrated an understanding of the control problem including limitations?	30%	
Design and Analytical skills: Have the models (wind turbine, batter bank, super capacitor and the integrated system) been appropriately designed/presented/analysed?	30%	
Engineering Judgement and Recommendations: Have the design results been appropriately interpreted and the conclusions well argued? Has the student present a clear and defensible recommendation to the client?	20%	
Total Marks	100%	

References:

1. L. Gelažanskas, A. Baranauskas, K.A.A. Gamage, M. Ažubalis, “Hybrid wind power balance control strategy using thermal power, hydro power and flow batteries”, International Journal of Electrical Power & Energy Systems, Volume 74, January 2016, Pages 310-321.
2. MATLAB, “Supercapacitor Parameter Identification”, accessed online on 02/10/2018 https://uk.mathworks.com/help/physmod/sps/examples/supercapacitor-parameter-identification.html
3. Zubieta, L. and R. Bonert. "Characterization of Double-Layer Capacitors for Power Electronics Applications." IEEE Transactions on Industry Applications, Vol. 36, No. 1, 2000, pp. 199-205.